

PARTICULAR SPECIFICATION P2C

APPENDIX P2C – Employer Information Requirements

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P2C.1 DEFINITIONS

As-Built	The record drawings and documentation defining deviation to the designed information occurring during construction at the end of the project.
As-Constructed Model	Also referred to as Construction Progress Model information during construction. The “As-Constructed” model and its appended documentation are continually updated as construction progresses.
AIM	Asset Information Model - Maintained information model used to manage, maintain and operate the asset after construction.
BIM Guide	A guide to the use of BIM published by the Building and Construction Authority (The Singapore BIM Guide – Version 2).
BIM Model	A digital representation of the physical and functional characteristics of the project. The model comprises of 3-dimensional graphical information, non-graphical information and documentation.
Building Information Modelling	A process of designing, constructing or operating a building or infrastructure asset using electronic object-oriented information.
Clash rendition	Representation of the native format model file, used specifically for spatial planning and coordination processes. This is to achieve clash avoidance through clash analysis.
Classification	The action or process of categorising, grouping or sorting elements according to shared qualities or characteristics.
Common Data Environment (CDE)	Single source of information for any given project, used to collect, manage and disseminate all relevant project artefacts (including 3D models, drawings, documents etc.) for multi-disciplinary teams in a managed process.
CSA	Civil, Structural and Architectural disciplines
Design Authoring	A process in which BIM software is used to develop a Building Information Model based on criteria that is important to the translation of the building’s design.

Electronic Document Management System (EDMS)	System for storing, retrieving, sharing and otherwise managing electronic documents in line with BS1192.
Handover Model	As defined in BCA VDC Guide, the version of the Model that changes ownership and authorship. (i.e. The Released Model of design intent that is handed over to the awarded Contractor for their visual reference and further development).
IFC	Issue For Construction
.ifc (File extension)	Industry Foundation Classes (.ifc) – A data model intended to describe building and construction industry data in a platform neutral and open file format.
Level of Development (LOD)	The “Level of Development” is the degree to which the elements graphical and non-graphical (attributes) has been thought through a particular stage.
Level Of Detail (LoD)	Description of graphical content of models at each of the stages (PAS 1192-2:2013)
Level of Information (LoI)	Description of non-graphical content of models at each of the stages (PAS 1192-2:2013)
Master Information Delivery Plan (MIDP)	Primary plan for when project information is to be prepared, by whom and using what protocols and SMP (standard methods & procedures), incorporating all relevant Task Information Delivery Plans (TIDP).
MEICA	Process Mechanical, Electrical, Instrumentation, Controls and Automation disciplines
MEP	Non-Process Mechanical, Electrical and Plumbing disciplines. Also referred to as Building Services)
OPM	Open Plant Modeler
OPPID	Open Plant P&ID
Project Information Model (PIM)	As defined in BS1192, the information model developed during the design and construction phase of a project. PIM comprises of a series of individual models, federated models and non-graphical information developed progressively during the project.

Released Model	As defined by BCA BIM Particular Conditions, this means a Model identified as such and released by the Model Author from time to time as agreed in the BIM Execution Plan.
Virtual Construction Model (VCM)	Subsequent version of the project information model developed from the VDM by the contractor and their supply chain.
Virtual Design Model (VDM)	This is the version of the Project Information Model (PIM) developed by the Consultant (for CSA) and Contractor (for MEP and MEICA elements).
4D	Construction Sequencing.
5D	Quantities and Cost Tracking.
6D	Asset Management.

P2C.2 INTRODUCTION

- (a) The Employer Information Requirements (EIR) sets out the Contractor's requirements under the relevant contract to deliver BIM at TWRP project. This document shall be read in conjunction with the following documents. Where any requirements are not explicitly specified in the Particular Specification or this EIR, the list of documents below should be referenced with following order:
- (i) Particular-Specifications;
 - (ii) This EIR;
 - (iii) Information Management Package;
 - (iv) G1.2 Information Management (IM);
 - (v) CORENET Singapore BIM e-submissions
[https://www.corenet.gov.sg/general/building-information-modeling-\(bim\)-e-submission.aspx](https://www.corenet.gov.sg/general/building-information-modeling-(bim)-e-submission.aspx);
 - (vi) BCA Singapore BIM Guide version 2, 2013;
 - (vii) BCA Singapore VDC Guide version 1, 2017;
 - (viii) BCA BIM Particular Conditions Version 2015;
 - (ix) BCA BIM Guide for Asset Information Delivery Version 2018;
 - (x) Singapore Standard CP 83-1 : 2004 (2015), Organisation and naming of CAD layers;

- (xi) Singapore Standard CP 83-2 : 2000 (2015), CAD symbols;
 - (xii) Singapore Standard CP 83-4 : 2001 (2015), CAD drafting conventions;
 - (xiii) Singapore Standard CP 83-5 : 2001 (2015), Colour and line type;
 - (xiv) BS-1192:2007 Collaborative production of architectural, engineering and construction information – Code of practice, 2007;
 - (xv) PAS 1192-2:2013 Specification for information management for the capital/delivery phase of construction projects using building information modelling, 2013; and
 - (xvi) PAS 1192-3:2014 Specification for information management for the operational phase of assets using building information modelling, 2014.
- (b) The Contractor shall read this EIR with Board's Information Requirements (BIR) in G1.2.
- (c) The BIR and EIR also apply to the Contractor's supply chain. The Contractor shall ensure that all procurement contracts include the EIR for the Project and that the supply chain provides information in the correct format.
- (d) The Contractor's BIM Execution Plan shall be their method statement in response to this EIR , BIR, and the Contract. Unless otherwise expressly provided in the Contract, the EIR does not change any contractual obligations or shift any risks of the parties in the Project which has been agreed in the Contracts.
- (e) Exceptions to any aspect of the BIR and EIR, shall be subject to the agreement of the S.O. and where agreed, are to be highlighted in the Contractor's BIM Execution Plan (BEP).

P2C.3 PROJECT BIM GOALS OBJECTIVES

- (a) The TWRP project will be delivered using Building Information Management (BIM) principles as referenced in G1.2 and the Singapore BIM standard when providing the Works. This document further clarifies BIR in G1.2, and is also referred to EIR in PAS 1192. The Contractor shall comply with the EIR and BIR described herein. The S.O. will maintain the EIR and shall communicate any changes or updates, and the Contractor shall comply as the Works progress.

- (b) The BIM principles to be followed for the TWRP project are based on BS 1192: 2007 and PAS 1192. An Information Model comprises Documentation, Graphical and Non-Graphical data which is enriched over time.
- (c) The main objectives of using BIM to deliver the Works and the high-level methodology for achieving each objective are given in Table P2C-1.

Table P2C-1: Project BIM Goals and Objectives

Goal	Objective
Enhance ability to make informed and timely decisions and communicate with stakeholders	Leverage model visualizations and intelligent data to support decision making processes and present information in a fashion that is easily understood.
Common working environment for the team to collaborate and succeed	Enhance collaboration and communication amongst all stakeholders to improve the quality of models and ensure timely project implementation.
Enhance worker productivity due to better coordinated information	Align to the mission of the Board and BCA to enhance productivity for all project participants to eliminate waste and deliver work efficiently.
Mitigate project risk to Board in the form of disconnected data and inaccurate models	Drive the process of information and model management to eliminate cost overruns at the project site where coordination is exponentially more expensive than the virtual environment.
Increase confidence and predictability of cost and schedule	Integrate intelligent model data with cost and scheduling activities. This will increase the ability to efficiently review costs for multiple options and understand potential schedule conflicts for complex construction activities.
Oversight of actual construction progress vs. virtually planned work	Simulate the complex construction sequence to identify and communicate any potential risks virtually, first.
Control and streamline information for seamless turnover and best in class Asset Management environment	Create a controlled data environment throughout the life cycle of the project to facilitate a digital exchange of operations and maintenance information at project completion. Mitigate the painful process of manual information input at the close of the programme. Reduce the risk of maintenance and operations items that may require early attention while information is being input.
Use new and innovative technologies where appropriate	Analyse the usefulness of innovative technologies that can be applied throughout the project to deliver planned business results.

P2C.4 MODELLING REQUIREMENTS

- (a) The Information Model will be the sole source from which 2D drawings, 3D models, 4D construction sequences, 5D quantities and cost tracking and 6D asset management deliverables are to be derived. The Contractor shall create models that are both complete and accurate, in accordance with this specification and with the Contractor's approved BIM Execution Plan.
 - (i) 2D denotes drawings in native dgn file and Adobe PDF. The Contractor shall provide soft copies of all drawing sheets for review/comment by the S.O..
 - (ii) 3D denotes 3D models created using the BIM Authoring tools required for the Works defined in Table P2C-3.
 - (iii) 4D denotes construction sequence/programme information. This process will require mapping 3D objects to the date/time as shown in the P6 Programme for the Works specified in the document G2.2.
 - (iv) 5D denotes quantification and project expenditure tracking. The Contractor shall provide model with quantification schedules attached specified in the document G2.2.
 - (v) 6D denotes asset management. Prior to completion of construction, the Contractor shall embed Construction Operations Building Information Exchange (COBie) and PUB's asset management data on all model elements that represent "Managed Assets". Such data shall be entirely suitable for incorporation into the Board's asset management system (AMS). The specific data to be compiled for turnover to the Board will be provided by the S.O. upon Contract award.
- (b) At the time of Tender, the S.O. has developed BIM models for civil, structural and MEICA work using the design/modelling software given in Table P2C-3. A set of modelling guidelines for BIM elements at different stages of the project are provided in the latest version of the Singapore BIM Guide (refer to <https://www.bca.gov.sg/bim/bimlinks.html>) and shall be referenced.
- (c) The S.O. will furnish the Contractor with the Handover Model; to meet the Programme for the Works. Once released, as stated in G1.2, the Contractor shall maintain and update the Handover model(s) as model author.
- (d) The Contractor shall develop the Handover Model and manage the PIM in accordance with the Master Information Delivery Plan (MIDP) within Contractor's CDE, and share with S.O. for periodic review and approval.
- (e) The Contractor shall use the outputs from BIM models for 2D and 3D drawing production, visualisations, simulation, project expenditure analysis, collaboration, planning and documentation throughout the project life cycle. These outputs shall include reports, extracted CAD Drawings and non-geometrical data including schedule/datasheet describing physical, functional and performance of the plant and equipment provided.

- (f) In general, each element shall be modelled according to its size, shape, location, orientation and quantity. At the tender stage of the project, element properties are necessarily approximate. The Contractor shall develop each element with accuracy as the project progresses, until the level of accuracy is sufficient to support construction and installation.
- (g) Architectural, Structural and Mechanical/Electrical/Plumbing (MEP) modelling guidelines and templates for Singapore BIM e-submissions for regulatory purposes can be found at: [https://www.corenet.gov.sg/general/building-information-modeling-\(bim\)-e-submission.aspx](https://www.corenet.gov.sg/general/building-information-modeling-(bim)-e-submission.aspx).
- (h) Drawings shall fully be derived and generated from BIM Models possible. Drawings shall include, but not be limited to plan and sectional views, elevations, profiles, quantities and schedules; and shall be delivered as representations of or extractions from BIM Models.
 - (i) 2D Drawings (typical details, notes, legends, etc.) not extracted from the BIM Models shall be clearly identified on drawing note section and in the Drawings List that all Project Teams will use to plan their document sets. The use of these 2D Drawings on the project shall be subject to the PM/S.O. review and approval prior to the creation of these 2D Drawings to ensure that BIM is being used to its fullest extent.
 - (ii) All 2D Drawings and Construction Documentation will adhere to, and comply with the Singapore CAD Standard CP83.
- (i) The Contractor shall be fully responsible for the following:
 - (i) For any scope or design change, the Contractor shall update the models. In structural design, S.O. will issue an instruction upon Authority's approval and the Contractor shall update the models with the instruction;
 - (ii) For elements of the Works for which the S.O. completes design, the Contractor is to maintain the BIM model and update the model with as-built data;
 - (iii) For elements of the Works for which the Contractor is to complete the design (e.g. MEICA), the Contractor shall update the model with approved designs and provide an as-built model along with the linked database upon completion;
 - (iv) Any temporary works which include but are not limited to vehicle and crane movements, temporary facilities, temporary roads, temporary works, formworks, scaffoldings those activities which impact any other Contract or require permits. A model may also be required to demonstrate that an activity is safe and/or does not impact other Contractors or within contracted work; and
 - (v) Model of temporary structure as per supplier shop drawings, details and method statements. Follow the BCA's BIM Essential guide for collaborative virtual design and construction.

- (j) The Contractor shall comply with the requirements of BCA for statutory submissions to regulatory agencies, as mandated by the prevailing BCA BIM e-submission regulations. The Contractor shall be responsible for liaison with BCA and other relevant regulatory agencies on all consultations and submissions related to the BIM development.
- (k) All models shall be compliant with the specifications. Taking the approach also that “if it is to be constructed it needs”.

P2C.5 LEVEL OF DEVELOPMENT

- (a) The Contractor shall follow Level of Development(LOD) as defined in the following documents unless specified by the S.O.:
 - (i) BIM Forum Level of Development Specifications (<https://bimforum.org/lof/>);
 - (ii) BCA Singapore BIM Guide version 2; and
 - (iii) BCA Code of Practice on Virtual Design and Construction (VDC).
- (b) The S.O. will produce LOD 300 information for CSA elements and LOD 200 for MEP and MEICA elements. The Contractor shall develop models to be minimum of LOD 400 for the PIM and LOD 500 for AIM.
- (c) A breakdown of the expected LOD is provided in Table P2C-2. This breakdown is non-exhaustive and provides the Board’s intent. The Contractor shall submit own LOD as a part of BEP by expanding on the Model Element Table to provide models at each stage to meet the LOD requirement.

Table P2C-2: Level of Development Per Milestone

Project Milestone	Conceptual Design/Feasibility Study	Preliminary Design	Detailed Design	Construction	As-built
LOD	LOD 100	LOD 200	LOD 300	LOD 400	LOD 500
CSA	PM	S.O.	S.O.	Contractor	Contractor
MEICA	PM	S.O.	Contractor	Contractor	Contractor

P2C.6 INFORMATION REQUIREMENTS

- (a) The Contractor shall utilise the defined non-graphical data content which shall include, but shall not be limited to the following:
 - (i) BS 1192 Status, Revision, Workflow;
 - (ii) Document numbering;

- (iii) Drawing numbering;
 - (iv) 4D sequence codes;
 - (v) 5D quantity take-off (QTO) codes;
 - (vi) 6D attributes;
 - (vii) Process-mechanical tagging codes;
 - (viii) Non- process-mechanical tagging codes;
 - (ix) Electrical tagging codes;
 - (x) Classification codes;
 - (xi) Interface lists;
 - (xii) Material/Service/Fluid Codes;
 - (xiii) Construction operations building information exchange (COBie); and
 - (xiv) Board's asset management system (AMS).
- (b) The codes referenced above will be made available to the Contractor upon award. Management of the codes shall remain with the S.O.. Any changes or amendments requested to any of the code lists shall be administered by the S.O..
- (c) Laser Scanning
- (i) Where specified within the Particular Specifications P2 Volume 3, the Contractor shall refer to the laser scanning requirements.
 - (ii) The contractor to provide the following as a minimum 6 monthly basis as part of the progress report documentation or as required by S.O.:
 - 1) If required by the S.O. the point cloud file of installed equipment or constructed permanent works elements during the reporting month. The scan is to be performed after the removal of any temporary works of that element;
 - 2) The BIM rendered progress scan point cloud file shall be indexed to be compatible with Bentley's point cloud engine (PointTools) for native import; and
 - 3) The BIM rendered progress scan point clouds shall be superimposed/overlay into both the Contractor VCM LOD 400(fabrication/shop) and 500 (as-constructed) model created with design/modelling software given in Table P2C-3 following the visual verification by the Contractor's site team (RE and RTO).

- (iii) A report of the alignment and/or deviation (delta) shall be included and presented in the bi-monthly BIM progress meeting.
 - (iv) The Contractor shall use the feature extraction feature in OPM for MEICA elements detected. A report shall be generated to be discussed during the bi-monthly progress meeting.
- (d) P&ID Data
 - (i) P&ID data is held in multiple relational databases and will be made available to the Contractor early in the Contract Period. The databases will be synchronised with P&ID drawings and shall be kept up to date by the Contractor with any Contractor changes. The P&ID database shall also be kept up to date with any Contractor-designed elements. The Contractor shall complete the database prior to handover of the asset.
 - (ii) The Contractor is expected to have suitably qualified persons to administer an SQL database and provide interrogation, issue resolution and exporting of content from SQL. The Contractor shall submit the P&ID database along with the As-Built Model.
 - (iii) P&ID information is a part of the LoI (held in databases) which will be provided by the S.O. upon contract award. The Contractor shall seek approval from the S.O. for any new elements to add into the database.
- (e) Integration with Asset Management System (AMS)
 - (i) The Board has set up an AMS for the whole organisation and, in addition to the BIM data, data shall be provided by the Contractor for AMS Integration.
 - (ii) The Contractor shall configure 6D parameters in their BIM authoring software and populate the parameters at the following suggested minimum data drops stages:
 - 1) 6D Data Drop 1: During asset/product detailed design or equipment requisition;
 - 2) 6D Data Drop 2: During asset delivery, installation or construction; and
 - 3) 6D Data Drop 3: During asset commissioning, startup and operation.
 - (iii) The following are the items to be included as a minimum in the data drops mentioned above (these will be further defined and confirmed upon contract award):
 - 1) 6D Data Drop 1: All details from final approved data sheet;
 - 2) 6D Data Drop 2: Additional installation details, FAT test sheets, cost, 4D and 5D codes; and
 - 3) 6D Data Drop 3: Maintenance task details, SAT test sheets, O&M training & instructions, HAC register, confined space register.

- (f) These deliverables and timing to be reflected in the MIDP as defined in Section 9 in order for the S.O. to review and verify BIM progress.
- (g) The specific requirements for the items listed for each data drop to be referenced within the specifications in the tender documentation.
- (h) The Contractor will be provided with a soft copy of the latest templates used by the Board, which will include, but not limited to, the asset master list (e.g. description, asset ID, runtime, type, location, model, serial number, brand, design capacity, cost, date, warranty, spares list, checklist, and maintenance schedules). An example of the existing templates are provided in the tender document for reference. The Contractor shall include in his rates and prices to populate and submit all the required information until successful integration to Board's AMS and to the satisfaction of the S.O..
- (i) The Contractor shall provide computerised asset listing data/information for data integration to the Board's AMS before end of the Contract.

P2C.7 COLLABORATION AND COORDINATION

- (a) All Information for coordination and collaboration should be managed in accordance with the processes described in BS1192 and PAS1192-2.
- (b) Common Data Environment (CDE) - The Contractor shall maintain a Common Data Environment (CDE), to store and manage all information relevant to the Contract, and the CDE shall.
 - (i) Serve as a single source of information used to share contract information with the S.O.;
 - (ii) Exchange information with the S.O.'s project wide CDE, referred to as the TWRP CDE. Information transfer shall be via the S.O.'s Project Management Information System (TWRP PMIS);
 - (iii) Include an electronic document management system (EDMS) for management of all documentation transfers among the Contractor and the supply chain;
 - (iv) Be protected against cyber-attacks through compliance with Multi-Tier Cloud Security Singapore standard level 3 for cloud based solutions. The Contractor shall define the tier to be applied and provide the necessary evidence of applicable tier in the BIM Execution Plan prior to implementation alongside the proposed solution;
 - (v) Include disaster recovery system, which shall allow data to be backed up at scheduled intervals to a secure server for recovery; and
 - (vi) Enforce BS-1192 Work Processes for the Contractor and the supply chain.

- (c) Model Management - All models will be managed, coordinated and developed in the Contractor CDE per applicable standards.
 - (i) The BIM model shall be created using the real-world coordinates, and shall comply to the Land Surveyors Board (LSB) Directive on Engineering and Hydrographic Survey Practices and the Standard and Specifications for 3D Topographic Surveying (Mapping) in Singapore (Version 1.0 – 2013) produced by the Singapore Land Authority (SLA). All project spatial information shall be referenced to the Singapore Transverse Mercator Grid (SVY 21 datum – 2006), with vertical control derived from the SGeoid09 - geometric geoid model established by SLA. Any GNSS survey work shall take reference to the Singapore Satellite Positioning Reference Network (SiReNT).
 - (ii) All Contractors/Sub-Contractors shall adhere to a File Naming Convention provided by the S.O. upon Contract award which will be based on BS 1192; and
 - (iii) The Contractor to follow the Grid rules outlines by S.O. when developing models and 2D drawings.
- (d) Coordination – The contractor shall use models for internal coordination including supply chain not limited to:
 - (i) Demonstration of the coordination of trades/disciplines through the models;
 - (ii) Coordination and control of the design and construction activities;
 - (iii) H&S use/verification; and
 - (iv) Interface management with other packages.
- (e) The Contractor shall arrange regular internal BIM meetings for coordination with own engineering team and sub-contractors. The contractor shall include a bi-weekly BIM Coordination meeting with S.O.. This meeting to be attended by the Contractor's BIM Manager and BIM leads, the S.O. BIM Manager and BIM Coordinator and any other personnel as required. The agenda at the meeting to include the following as a minimum:
 - (i) Review S.O. comments of the submitted monthly model files;
 - (ii) Status of the Contractor's BIM deliverables;
 - (iii) Contractor's clash detection reports; and
 - (iv) Construction sequencing (4D simulation).

(f) Clash Detection

- (i) The Contractor to provide clash detection reports with “Zero Clash” for the following model groupings to the S.O. for approval prior to construction:
 - 1) Architectural versus Structural;
 - 2) (Architectural + Structural) versus Civil;
 - 3) (Architectural + Structural) versus (Building Service);
 - 4) (Architectural + Structural) versus (MEICA); and
 - 5) Building Services versus MEICA .
- (ii) The minimum requirements for coordination and clash detection shall be defined in the Clash Resolution Matrix with soft and hard clash detection. The models shall be used to verify that clashes are resolved at each stage of the project by the contractor.
- (iii) Processes and Tolerances for clash detection shall be detailed in the BEP.

(g) Information Exchange

- (i) All contained files shall be named to the file naming standard defined by S.O..
- (ii) Exchanged files will be verified by the file name as part of the CDE upload or receipt process. Only correctly named files will be accepted. Correctly named files will be uploaded/accepted by the CDE without further profiling into the correct folder structure position.
- (iii) Frequency of sharing the models shall be on a weekly basis between disciplines/trades, subcontractors and other parties in the supply chain as necessary. Federated models shall be shared and published on the monthly basis to S.O. (in addition to the requirements of the project stages).
- (iv) Published files will validated against the model and information requirement before exchanged.

(h) Model and Drawing Change Management

- (i) The Contractor shall follow own change management process to manage the models and drawings.
- (ii) The Contractor shall update the Change log registers and submit to S.O. with every submitted revision of the models and drawings. The Change log registers template will be provided upon contract award. The Contractor to note the following:
 - 1) Change log register to be maintained for each Model / Drawing updates;

- 2) Change log register to be maintained by Model Author;
- 3) Models and drawings shall be updated per TWRP Contractor's common metadata dictionary for the revision codes to be used at every submittal;
- 4) Compare the Contractor's change log register with S.O monthly basis.

P2C.8 BIM AUTHORIZING TOOLS

- (a) BIM Authoring Tools - Design/modelling Software to be used by the Contractor to deliver the Works shall be as described in Table P2C-3.

Table P2C-3: TWRP BIM Authoring Tools

Design/Modelling	Version	Function
Primavera	P6	Construction Scheduling
Microsoft Office	2016	All documentation
SQL Server	2019	Hosting P&ID database
Bentley OpenBuildings Designer	10.08.XX	Architectural, Structural + Design review/Clash Detection + 5D QTO + 6D COBie/AMS
Bentley OpenBuildings Designer	10.08.XX	MVAC, Electrical, Plumbing, Fire Protection + Design review/Clash Detection + 5D QTO + +6D COBie/AMS
Bentley OpenPlant Modeler	10.09.XX	Process Mechanical, Instrumentation & Controls + 6D AMS
Bentley Open Plant P&ID	V8i SS10	P&ID - Process Mechanical, Instrumentation & Controls + 6D AMS
Bentley OpenRoads Designer	10.09.XX	Civil - Site Grading, Roads, Tunnel, Utilities, Geotech and Site Analysis
Bentley View	10.15.XX	2D /3D Review
Bentley Navigator	10.02.01.XX	3D Design Collaboration
Bentley SYNCHRO 4D Pro	06.03.02	4D Sequencing
Adobe Acrobat	11.0+	2D Design Collaboration/Redlining
Autodesk NavisWorks	2017	Delivery of model to the Board/PM

- (b) Alternative software for each of the above functions shall NOT be used. Software for additional functions such as design verification, reinforcement design etc. shall be proposed by the Contractor for the approval of the S.O..
- (c) Throughout the project it is understood that software version will change. The S.O. will evaluate newer version and determine if software version can be upgraded to take advantage of added features, and or functionality that might increase efficiencies for the project.
- (d) Approved upgrades shall take place on a Project-wide level. There shall be no multiple versions of software used on the Project.

P2C.9 BIM EXECUTION PLAN

- (a) The Contractor shall provide a fully-detailed BIM Execution Plan (BEP) for execution of the Works. The BEP shall describe the process in which BIM will be implemented throughout the Project lifecycle in compliance with this specification.
- (b) Upon Contract award, the BIM Manager shall be responsible for the contract specific BIM Execution Plan, that adheres to the BIM Execution Plan Template provided by the S.O., adding (if any) the Contract unique BIM requirements for the awarded contract. This BIM Execution Plan will include feedback provided by the BIM Coordinator of each Discipline and shall be submitted to the S.O. within four (4) weeks of Contract award.
- (c) The BEP shall be updated and implemented in accordance with the requirements detailed herein. As the project progresses, any modifications to the objectives, project members, Model Authors, Model Users or basic BIM/CAD deliverables shall be agreed with the S.O. and documented in the BEP.
- (d) The BEP shall include the following information as a minimum:
 - (i) Project information - general contract scope/ summary information, the project team, key personnel contact details related with BIM and Information Management;
 - (ii) BIM objectives and goals - BIM/CAD deliverables shall include models that describe the physical, functional and performance characteristics of the Works for visualization, simulation, analysis, collaboration, planning and documentation throughout the project lifecycle;
 - (iii) BIM Uses – The Contractor shall define own BIM uses for the model and methodology to achieve those uses. These shall include but not be limited to Construction Sequencing (4D), Cost estimation (5D), Asset Management data turnover (6D);
 - (iv) Governance, roles and responsibilities - the Contractor shall identify in each BEP, information management roles as outlined in PAS 1192;

- (v) BIM protocols - The Contractor shall identify and record, in each BEP, the BIM protocols used to create, manage and deliver models and information to meet the Particular-Specification, EIR and BIR;
- (vi) Model development approach – the model shall be progressively developed by the Contractor as the Works progresses. The S.O.'s tender model shall be developed to become a virtual construction model, and then as an as-built model which will be handed over to the Board at the time of Substantial Completion;
- (vii) Level of Development - The Contractor shall develop own LOD and confirm the models that will be delivered with Level of Detail and Level of Information;
- (viii) Procedures, methods, collaboration and training - the Contractor shall describe the methods, procedures and processes related to the BIM uses, protocols, collaboration and trainings;
- (ix) Supply chain – Define how the contractor will engage their suppliers to ensure that they are fully engaging with the Contract;
- (x) Master information delivery plans (MIDP) - The Contractor to submit MIDP for each submission identified in the Programme for the Works, including a detailed information delivery schedule and register containing all identified information deliverables, coordination items and any supporting data or information. The Contractor to submit the updated MIDP on a monthly basis with the Progress model submission;
- (xi) Interface management – Define how the contractor will use the model to help interfacing with other contractors;
- (xii) BIM compliance plan - The BIM/CAD deliverables shall meet the quality assurance guidelines as stated in Quality Control requirements provided by the PM/S.O. upon Contract award. Any deviation of quality assurance guidelines shall be outlined in the BIM Execution Plan for the PM/S.O.'s review and acceptance. The contractor shall detail model and data compliance will be detailed within the BIM Execution plan (BEP) to include, but not limited to:
 - 1) Quality assurance/control procedure; and
 - 2) Level of assurance on BIM requirements;
- (e) The content of each of these sections, and all other necessary BEP sections, shall be developed by the Contractor and shall be subject to the approval of the S.O..

P2C.10 SUBMISSION REQUIREMENTS

- (a) The contractor shall make submissions based on Master Information Delivery Plan (MIDP) to manage the delivery of information during the contract period. The BIM progress payment will be based on the completion of BIM deliverables in accordance with the forecast shown in the MIDP. The forecast dates should be aligned with the relevant task within the Programme for the Works.
- (b) Model Submissions during Design and Engineering
 - (i) 3D - the updated native model including all approved designs, all approved changes;
 - (ii) Clash resolution including clash detection reports; and
 - (iii) A completed preparation and validation checklist for each model that the Contractor shall develop and provide.
- (c) Model Submissions during Construction
 - (i) The Contractor shall provide information updates every month during execution of the Works and to be submitted every 8th of the month for S.O. review and approval. This shall include (as appropriate as the Works progress) models including:
 - 1) 3D - the updated native model including all approved designs, all construction works completed or in progress, all approved changes;
 - 2) 4D - the model(s) with embedded Programme for the Works. The model shall include temporary works, planned permitting works, interfaces with any other contractors as a minimum;
 - 3) 5D - the model(s) with associated exports of data for project expenditure tracking and for verification of monthly progress claims;
 - 4) 6D – updated models with as-built information embedded for each section of work completed in accordance with relevant data drop stage;
 - 5) Clash resolution including clash detection reports;
 - 6) Laser scanning point cloud file, and visual verification report;
 - 7) A completed preparation and validation checklist for each model that the Contractor shall develop and provide;
 - 8) Supporting CAD (2D) information or as agreed in the BEP; and
 - 9) Supporting or other required documentation in PDF format or as agreed in the BEP.

- (d) Model Submissions After Construction (As-Built)
- (i) The Contractor shall develop As-built models in accordance with the Contract, corresponding also to the final approved amendment plans submitted to the relevant authorities.
 - (ii) As-Built BIM Models and Drawings shall be endorsed “As-Built” and one set of original plotted drawings and submitted together with their soft copy in DVD-R or portable storage media. All As-Built Models and Drawings in both hard and soft copies shall be submitted prior to Substantial Completion of the Works.
 - (iii) The Contractor shall prepare and submit two complete sets of As-Built Models and Drawings. Each complete set shall comprise the following:
 - 1) 1 copy of “Model and Drawing List Details” in excel format;
 - 2) 1 soft copy of As-Built BIM Models (Native File Format) updated with S.O approved contractor’s final design and embedded with COBie/AMS data to all elements identified by the Board as Facility Management (FM) and Operation & Maintenance items;
 - 3) COBie exported Spreadsheets in excel format;
 - 4) AMS exported Spreadsheets in excel format;
 - 5) Updated database of P&ID data from OPPID;
 - 6) Customized datasets and project configuration setup for BIM Authoring tools;
 - 7) 1 set of A1 size paper prints (drawings for statutory submissions shall be endorsed by the appropriate Professional Engineer, Singapore);
 - 8) 1 ser of Laser scanning point cloud file, and visual verification report;
 - 9) 1 soft copy of the As-Built Drawings in native “.dgn” format;
 - 10) 1 soft copy of the As-Built Model in “.i.dgn” format;
 - 11) 1 soft copy of the As-Built Model in “.bim” format; and
 - 12) 1 soft copy of the As-Built Model in Navisworks “.NWD” format.
- (e) As-Built BIM Models and Drawings will undergo a full review by the S.O. to ensure the As-Built data meets all requirements. Upon acceptance/approval, the As-Built BIM Models/Drawings and their storage media shall become the sole property of the Board. All related costs associated with complying with these requirements is deemed to have been allowed for in the tender, whether indicated explicitly or otherwise.

- (f) The Contractor shall provide following deliverable formats for any submission:
- (i) Native BIM Models
 - 1) All TWRP facility Models (".dgn");
 - 2) Point cloud files;
 - 3) BIM rendered scan files.
 - (ii) 2D
 - 1) 2D Drawing files (including native ".dgn" of all contracts drawings);
 - 2) 2D Adobe PDF (.pdf).
 - (iii) 3D Federated Model Files
 - 1) One Federated Model ".dgn", ".i.dgn," , and ".bim" of the respective contract scope with all linked information included;
 - 2) Facility level Federated models ".dgn", ".i.dgn," of the respective with all linked information included;
 - 3) Navisworks ".nwd" with all linked information included.
 - (iv) 4D
 - 1) Bentley SYNCHRO 4D Pro : ".sp";
 - 2) Navisworks : ".nwd" and ".nwf";
 - 3) Primavera P6 : ".xer" or ".xlsx".
 - (v) 5D
 - 1) 5D Code embedded in native model files.
 - 2) BoQ report from models for project expenditure : ".xlsx" format.
 - (vi) 6D
 - 1) BIM Model with COBie/AMS data in ".ifc2x3";
 - 2) COBie (.xlsx format) workbook; and
 - 3) Asset Management System(AMS) – as per the spreadsheet (".xlsx" format).
 - (vii) OPPID Database
 - (viii) Any customized datasets and project configuration setup for BIM Authoring tools.

P2C.11 APPENDICES

- (a) **Information Management package – will be provided after the Contract award.**

Sl. No.	File Name	Description
Information/Document Control		
1	131101-WGS00-RG-IM-0008	TWRP Contractors Common Metadata Dictionary
2	131101-WGS00-RG-IM-0009	TWRP Contractors Document Naming Guidelines
CAD Standards		
3	131101-WGS00-DR-Z-0001	Title Block Template
4	131101-WGS00-PP-IM-0103	CAD Graphics Standard
5	131101-WGS00-M3-Z-0001	TWRP - Annotation, Legends, Symbol & Callout
6	131101-WGS00-FM-IM-0008	Drawing Sheet Review Checklist Template
7	131101-WGS00-PP-IM-0104	Redline Markup Procedure
BIM		
8	131101-WGS00-FM-IM-0001	BIM Execution Plan Template
9	131101-WGS00-PP-IM-0101	BIM Deliverable Matrix
10	131101-WGS00-FM-IM-0002	TWRP MIDP Template
11	131101-WGS00-PP-IM-0102	BIM Element LOD
12	131101-WGS00-FM-IM-0004	Model and Drawing Change Log Register Template
13	131101-WGS00-PP-IM-0105	BIM Model QC Check List Guidelines
14	131101-WGS00-FM-IM-0005	TWRP Clash Detection Report Template
15	131101-WGS00-FM-IM-0006	Model Review Checklist Template
16	131101-WGS00-FM-IM-0007	Federated Model Checklist Template
17	131101-WGS00-PP-IM-0110	Federated Model View Setup
AIM		
18	131101-WGS00-PP-IM-0109	Cobie Attributes - Mech
19	131101-WGS00-PP-IM-0108	Asset Management System Parameters
20	131101-WGS00-FM-IM-0009	Asset Data Template
Software Dataset		
21	131101-WGS00-PP-IM-0106	TWRP_Project Setup Procedure -OBD and OPM
22	131101-WGS00-PP-IM-0107	TWRP_Project Setup procedure -OPPID
23	131101-WGS00-PP-IM-0112	TWRP Bentley CONNECT Version Migration guide – OBD and OPM
24	131101-WGS00-PP-IM-0111	TWRP Dataset Change Management
25	131101-WGS00-FM-IM-0012	TWRP Dataset Change Request Form

END OF SECTION